

Hermes Architecture Evolution — Website Maintenance as an Engineering Capability (v0.3)

Background

A key design question emerged during the evolution of Hermes:

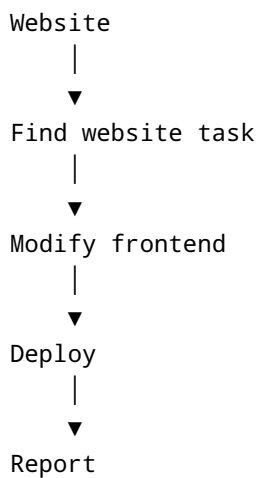
Can an Autonomous Website Maintenance Agent still be part of Hermes?

The answer is **yes**, but only if it is positioned correctly within the architecture.

The problem was never the existence of a Website Maintenance Agent.

The problem was allowing it to become the primary objective of the system.

Original Architecture (Problem)



Characteristics:

- Website becomes the mission.
- Engineering exists only to improve the website.
- Reports describe website work.
- Engineering knowledge is limited.
- Little measurable value is produced.

This architecture naturally optimizes for:

- commits
- deployments
- visual improvements

- appearance of productivity

instead of engineering outcomes.

Correct Architecture

The hierarchy should instead be:

```
Mission
|
Objectives
|
Engineering Work
|
Knowledge
|
Documentation
|
Website
```

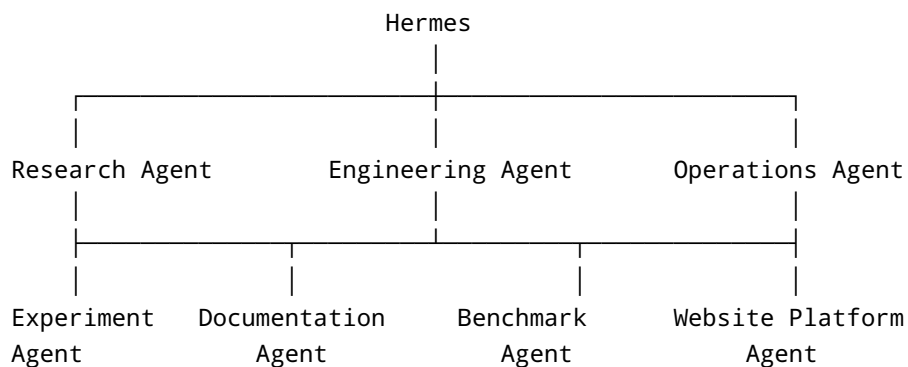
The website becomes an output.

Not the objective.

Hermes is an Engineering Organization

Hermes should be viewed as a virtual engineering organization composed of multiple specialized agents.

Example:



Every agent owns a different engineering responsibility.

No single agent defines the mission.

Website Platform Agent

Rather than treating website maintenance as the mission, it becomes one engineering capability inside Hermes.

Responsibilities include:

- UI improvements
- Accessibility
- SEO
- Performance optimization
- Bundle size reduction
- Responsive design
- Dependency management
- Broken link detection
- Navigation improvements
- Technical health of the website

These are legitimate engineering activities.

They simply should not dominate Hermes' priorities.

Engineering Flow

Instead of:

Website

↓

Need content

↓

Create engineering task

Hermes should operate as:

Engineering Question

↓

Research

↓

Experiment

↓

Measurement

↓

Decision

↓

Documentation

↓

Website Publication

The website consumes engineering outputs.

It does not generate engineering objectives.

Example Workflow

Research Agent asks:

Can Model A replace Model B for code review?

↓

Experiment Agent

Runs benchmark.

↓

Benchmark Agent

Measures:

- latency
- accuracy
- cost
- reliability

↓

Engineering Agent

Makes recommendation.

↓

Documentation Agent

Produces engineering report.

↓

Website Platform Agent

Updates:

- experiment registry
- dashboard
- metrics
- architecture notes
- public documentation

The Website Platform Agent publishes engineering work rather than inventing it.

Website Agent Backlog

A Website Platform Agent should maintain its own engineering backlog.

Examples:

- Improve Lighthouse score
- Improve accessibility
- Reduce JavaScript bundle size
- Optimize image loading
- Improve SEO
- Improve documentation navigation
- Update dependencies
- Fix broken links
- Improve analytics
- Improve responsive layout

These tasks remain valuable.

They simply compete alongside other engineering priorities rather than replacing them.

Engineering Categories

Hermes should classify work into multiple categories.

Core Engineering

Examples:

- AI agents
- infrastructure
- APIs
- automation
- production software

Priority:

Highest

Experiments

Examples:

- model benchmarking
- architecture validation
- AI evaluation
- retrieval experiments

Priority:

High

Operations

Examples:

- CI/CD
- deployment
- monitoring
- backups
- observability

Priority:

High

Documentation

Examples:

- engineering reports
- ADRs
- technical notes
- tutorials

Priority:

Medium

Website Platform

Examples:

- frontend
- accessibility
- SEO
- performance
- documentation experience

Priority:

Medium

Technical Debt

Examples:

- refactoring
- cleanup
- dependency updates

Priority:

Medium

Exploration

Examples:

- papers
- prototypes
- new tooling

- emerging technologies

Priority:

Low

Event-Driven Website

Rather than asking the Website Platform Agent to constantly search for website improvements, it should react to engineering events.

Example:

Experiment Completed

↓

Publish experiment page

Architecture Decision

↓

Update ADR section

Benchmark Finished

↓

Update dashboard metrics

Research Completed

↓

Publish engineering article

New Release

↓

Update changelog

The website becomes a live reflection of Hermes' engineering work.

Naming Considerations

The name "Autonomous Website Maintenance Agent" is technically correct if the scope is limited to maintaining the web application.

Alternative names may better communicate broader responsibilities.

Possible names include:

- Website Platform Agent
- Publishing Agent
- Documentation Experience Agent
- Knowledge Portal Agent

The preferred name depends on whether the agent only maintains the website or also manages publication, documentation, search, navigation, and knowledge presentation.

Architectural Principle

Hermes owns the mission.

Individual agents own subsystems.

The Website Platform Agent is one subsystem among many.

It does not determine engineering priorities.

It receives engineering outputs and ensures they are presented effectively to users.

Final Principle

The website should always be a consequence of engineering work.

Never the justification for engineering work.

If Hermes stops running tomorrow, tabaco.id should remain a valuable engineering knowledge base because it documents real experiments, real software, real decisions, real metrics, and real lessons learned.

That is the distinction between a self-maintaining website and a true autonomous engineering laboratory.